

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

INTEROFFICE COMMUNICATION

TO: Daniel Dell, Permits Section, Water Bureau

FROM: Doyle Brunsen, Surface Water Assessment Section, Water Bureau

DATE: January 31, 2008

SUBJECT: Cadillac Wastewater Treatment Plant (WWTP)
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257
Water Quality-Based Effluent Limits (WQBEL)

We have completed our evaluation for WQBELs for the reissuance of the NPDES permit for the Cadillac WWTP. They continuously discharge 3.2 million gallons per day (MGD) (4.93 cubic feet per second [cfs]) of treated municipal wastewater to the Clam River (NE 1/4, NE 1/4, Section 33, R9W, T22N) in Wexford County.

The Clam River is protected for warmwater fish from Lake Cadillac to County Line Road from May – September, and coldwater fish the remainder of the year as well as other indigenous aquatic life and wildlife, fish consumption, partial body contact recreation, total body contact recreation from May through October, agriculture, navigation, industrial water supply, and public water supply at the point of intake.

The monthly 50% and 95% exceedance flows, in cfs, for the Clam River at this location are as follows:

	<u>JAN.</u>	<u>FEB.</u>	<u>MAR.</u>	<u>APR.</u>	<u>MAY</u>	<u>JUNE</u>
95%	5.7	4.9	8.3	15	6.1	3.6
50%	17	15	37	67	31	9.9
	<u>JULY</u>	<u>AUG.</u>	<u>SEP.</u>	<u>OCT.</u>	<u>NOV.</u>	<u>DEC.</u>
95%	2.7	2.9	3.0	3.9	5.6	5.8
50%	6.6	5.8	6.7	14	20	21

These flows are updated from the previous flows used for recommendations.

RECOMMENDATIONS

The recommended limits are found in table 1. These recommendations meet the coldwater dissolved oxygen (D.O.) water quality standard (WQS) of 7 milligrams per liter (mg/l) as a daily minimum. Ammonia toxicity WQBELs are based on the 0.029 mg/l un-ionized ammonia toxicity criterion for coldwater streams.

We evaluated the effects of the discharge on the Clam River using a Multi-reach Streeter-Phelps D.O. model. We modeled the effect of the discharge on the receiving stream based on studies (time of passage, reaeration, flow measurements) done in 1983 (Synk, MDEQ Report #025120). The diurnal variation (daily average minus daily minimum concentrations) of 0.75 mg/l and the background D.O. are based on site specific data collected in 2006 (Brunsen, MI/DEQ/WB-08/Draft).

The recommendations contained herein are based on WQS. Treatment methods or economic concerns have not been addressed. We do not imply in these recommendations that other methods or considerations should not be used in deciding the permit limits.

db:rm

cc: Michael Stifler, Cadillac District Supervisor, Water Bureau
Brenda Sayles /Effluent Limit File, Water Bureau

TABLE 1
January 31, 2008

FACILITY: Cadillac WWTP
DISCHARGE: Continuous
FACILITY DESIGN FLOW: 3.2 MGD (4.93 cfs)
RECEIVING STREAM: Clam River

EFFLUENT					
PARAMETER	TIME PERIOD	CONC mg/l	MASS lb/d	COMMENT	RATIONALE
D.O.	MAY 1-SEP 30	6.0		DAILY MINIMUM	AWT
	OCT 1-NOV 30	6.0		DAILY MINIMUM	D.O. STANDARD
	DEC 1-APR 30	5.0		DAILY MINIMUM	D.O. STANDARD
CBOD ₅	MAY 1-NOV 30	10	270	DAILY MAXIMUM	AWT
	MAY 1-NOV 30	4	110	30-DAY AVERAGE	AWT
	DEC 1-MAR 31	41	1,100	DAILY MAXIMUM	D.O. STANDARD
	APR 1-APR 30	25	670	DAILY MAXIMUM	D.O. STANDARD
NH ₃ -N	MAY 1-NOV 30	2	53	DAILY MAXIMUM	AWT
	MAY 1-NOV 30	0.5	13	30-DAY AVERAGE	AWT
	DEC 1-APR 30	11	290	30-DAY AVERAGE	CHRONIC TOXICITY